# Scalability and performance Challenges in Bitcoin Blockchain Systems

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**ABSTRACT:**

**Background/Introduction**
Bitcoin blockchain is a digital system that enables the transfer of value without the need for a central authority, like a bank to facilitate the process. The system works by creating a block of transactions, which is then shared publicly on a network of computers. Even though this system is highly secure, transparent, and trustworthy, it has some limitations when it comes to the processing of transactions. The main problem that the system is facing is that, as the number of people using the system grows, the network is not efficient in the processing of transactions, leading to major problem known as scalability**.**

**Objectives**
The aim of this paper is to identify the scalability and performance issues faced by the Bitcoin blockchain system. It also tries to identify the major limitations faced by the system, such as low transactional throughput, block size limitations, and block generation delay. It also tries to evaluate the latest technologies like Segregated Witness and the Lightning Network to identify the advantages of using these technologies in the system.

**Methods/methodology**
This research is based on the qualitative study of existing literature, research papers, technical reports, and real-life scenarios related to Bitcoin and Blockchain technology scalability. Secondary data has been used in this research for studying the architecture, transaction processing mechanism, and limitations of Bitcoin technology. This paper also includes the study of Layer 2 solutions and protocol upgrades for checking their efficiency in terms of scalability.

**Results/Findings**
The result show that the scalability of the Bitcoin is major problem, as the block size is limited, as is the block generation time, leading to low throughput in the transactions. However, the problem is addressed by the Seg Wit, which optimises the block size by optimising the data in transactions, while the Lightning Network helps in the reduction of the network congestion by the off-chain transactions.

**Conclusion**
One of the most important concerns for Bitcoin technology is scalability. It has been seen that although the original design has focused on security and decentralisation, it has been lacking in terms of scalability. It has been observed that the implementation of Seg Wit and the Lightning Network has been effective in terms of scalability and efficiency, and it is expected that further development will be made in this regard.

**Keywords**: Bitcoin Scalability, SegWit, Lightning Network, Transaction Throughput, Decentralisation

## 1. INTRODUCTION:

Blockchain technology has become one of the most discussed innovations in computer science and digital system over the last decade. It provides a decentralised way of recording transactions where no central authority is required to control or verify the data. The idea was first introduced in 2008 by Satoshi Nakamoto through bitcoin. The main goal was to create a secure peer-to-peer digital payment system that does not depend on banks or intermediaries.

In simple terms, a blockchain is distributed ledger maintained by multiple nodes in a network. Each block stores transactions data along with a cryptographic hash of the previous block, which links all blocks together in a chain-like structure. Because of this design, once data is added to the blockchain, it becomes extremely difficult to modify. This ensures transparency, security, and trust among participants.

After bitcoin, blockchain technology evolved further with platforms such as Ethereum, which introduced smart contracts. These smart contracts allow developers to build decentralised applications (dApps) that can automatically execute agreements without human intervention. Due to this advancement, blockchain is now used in mainly areas such as finance, healthcare, supply chain management, and digital identity systems.

However, despite its advantages, blockchain system—especially public blockchains—face serious scalability and Performance challenges. For example, Bitcoin can process only around 7 transactions per second with an average block time 10 minutes. In comparison, traditional centralised system can handle thousands of transactions per second. As the number of users grows, public blockchain networks experience congestion, slower confirmations, and higher transaction fees.

The main reason behind these limitations includes fixed block sizes, consensus mechanism like Proof of Work that require heavy computational power, transaction validation overhead, and continuous growth of stored data (often referred to as the state bloat problem). These issues reduce the efficiency of blockchain systems and limit their use in large-scale real-world applications.

Therefore, improving scalability and performance while maintaining security and decentralisation has become one of the most important research challenges in blockchain technology today.

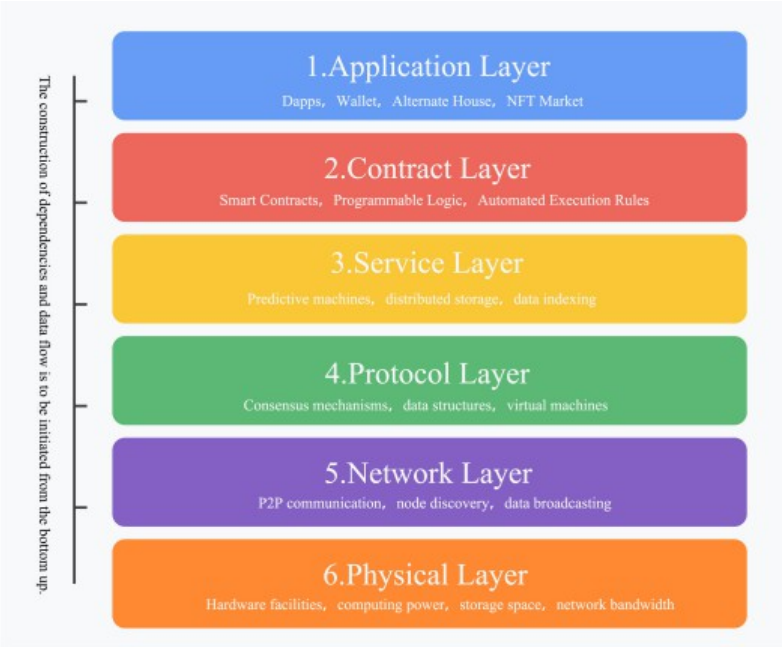



**Figure 4.** Blockchain layered architecture diagram.

## 2. BACKGROUND OF BLOCKCHAIN:

Blockchain scalability and performance have been widely discussed in recent years as public B324 blockchain networks continue to grow. The original design of Bitcoin introduced a secure and decentralised system using the proof of Work (PoW) consensus mechanism. While this design successfully ensured security and resistance against attacks, it also imposed certain limitations. Bitcoin maintains a fixed block size and an average block time of approximately of 10 minutes,

Which significantly restricts the number of transactions processed per second.

Researchers have pointed out that although this conservative design improves stability and security. It reduces transaction throughout and increases confirmation delays during high network activity.

With the introduction of Ethereum, Blockchain systems evolved beyond simple digital payments.

Ethereum enabled the execution of smart contracts and decentralised applications (dApps), increasing computational flexibility. However, this flexibility introduced additional complexity. The use of gas fees to measure computational cost helps prevent network abuse, but during periods of high demand, congestion leads to increased transaction fees and slower processing times. Several studies highlight that supporting smart contracts increases the computational load on nodes, which affects scalability and overall performance.

To address these issues, researchers have proposed multiple scalability solutions. One approach involves increasing block size to allow more transactions per block. While this method improves throughput, it may lead to centralization because larger blocks require more storage and bandwidth, limiting participation to high-capacity nodes. Another proposed solution is the implementation of Layer 2 technologies, such as off-chain payment channels, where transactions are processed outside the main blockchain and later settled on-chain. These techniques reduce on-chain congestion but introduce additional architectural complexity.

Sharding has also been suggested as a scalability solution. In this approach, the blockchain network is divided into smaller partitions called shards, each responsible for processing a subset of transactions. This parallel processing model increases transaction throughput but raises concerns regarding cross-shard communication and security consistency. Similarly, transitioning from Proof of Work to Proof of Stake (Pos) has been proposed to reduce computational overhead and energy consumption. While PoS improves efficiency, it introduces different security assumptions and governance challenges.

Another growing concern discussed in recent research is the state bloat problem. As blockchain networks grow, the amount of stored data continuously increases. Each full node must store and validate an expanding dataset, which increases storage requirements and synchronization time. This growth can discourage new nodes from joining the network, ultimately affecting decentralization.

Although significant progress has been made in proposing scalability solutions, most existing approaches involve trade-offs between security, decentralization, and performance. Many techniques either require substantial modifications to the existing protocol or introduce new layers of complexity. Therefore, there remains a need for optimization strategies that enhance performance while preserving the fundamental properties of blockchain systems.

## 3. SCALABILITY ISSUES IN BITCOIN SYSTEM

The scalability challenges observed in blockchain system are not accidental; they arise from several fundamental architectural and technical design decisions. Theses design choices were originally made to ensure strong **security, transparency, and decentralization** within the network. However, while these properties make blockchain system trustworthy and resistant to manipulation, they also introduce certain limitations that affect the system's ability to process a large number of transactions efficiently.

In case of bitcoin, multiple factors contribute to the scalability problem. Some of the most significant cause include the **block size limitation, block creation time, consensus mechanism overhead, sequential transaction processing, and network propagation delay.** Each of these factors influences the overall performance of the blockchain network.

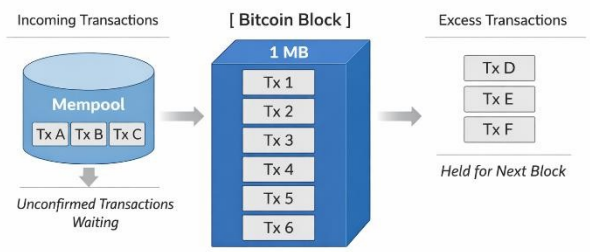
## 3.1. Block Size Limit

One of the most widely discussed cause of the scalability problem in the Bitcoin network is the **block size limitation.**

In Bitcoin, each block in the blockchain has a maximum size of approximately 1 megabyte (MB). Since every block contains a set of transactions, this size restriction directly limits the number of transactions that can be included within a single block.

Each transaction occupies a certain amount of space depending on its complexity and the number of inputs and outputs involved. Because of the fixed block size, only a limited number of transactions can fit inside a block. When the number of transactions submitted to the network becomes higher than the block capacity, the remaining transactions cannot be processed immediately.

**Figure 1:** Block Size Limitation and Transaction Capacity in Bitcoin




Instead, these transactions are placed in a waiting area known as memory pool (mempool) until they are selected by miners and included in future blocks. During periods of heavy network usage, the mempool can grow significantly, causing delays in transaction confirmations.

As results, the block size limitation directly restricts the transaction throughput of the bitcoin network, making it difficult for the system to support a very large number of users simultaneously.

## 3.2.Block Creation Time

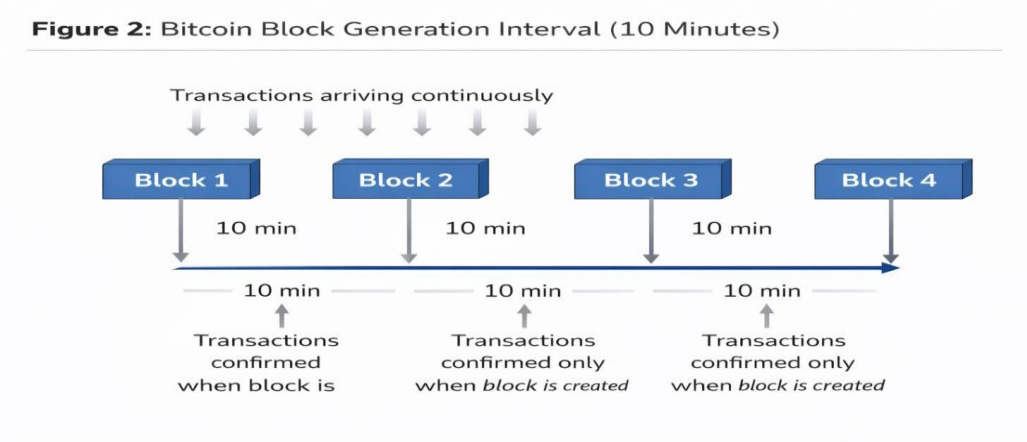
Another important factor that affects the scalability of the Bitcoin network is the block creation interval, also known as the block generation time. In bitcoin, a new block is created every 10 minutes. This time interval was intestinally designed by the system's creator to maintain network stability and allow sufficient time for newly generated blocks to propagate throughout the entire network.

Although this design improves reliability and reduces the chance of conflicts between miners, it also limits the speed at which new transactions can be added to the blockchain.

Since transactions can only be confirmed when they are included in a block, the relatively long block creation time leads to **slow transaction confirmation.** In many cases, users must wait several blocks before their

transaction is consider fully confirmed, which can take **30 to 60 minutes and even longer** during periods of high network congestion.

Therefore, the long block creation interval significantly restricts the rate at which transactions can be processed, contributing to the overall scalability problem.




## 3.3. Consensus Mechanism Overhead

However, Bitcoin Uses a consensus algorithm called proof of work (PoW), in which miners compete with one another to solve complex computational problems in order to add blocks to the blockchain.

Although Proof of Work is highly secure, it is computationally intensive and energy-intensive. The process of mining not only adds extra time to block creation but also to transaction confirmation.

Further, it is important to note that all nodes in the network have to validate the transaction and block before it is added to the blockchain. This again adds to the computational overhead and hence restricts the performance of the system.

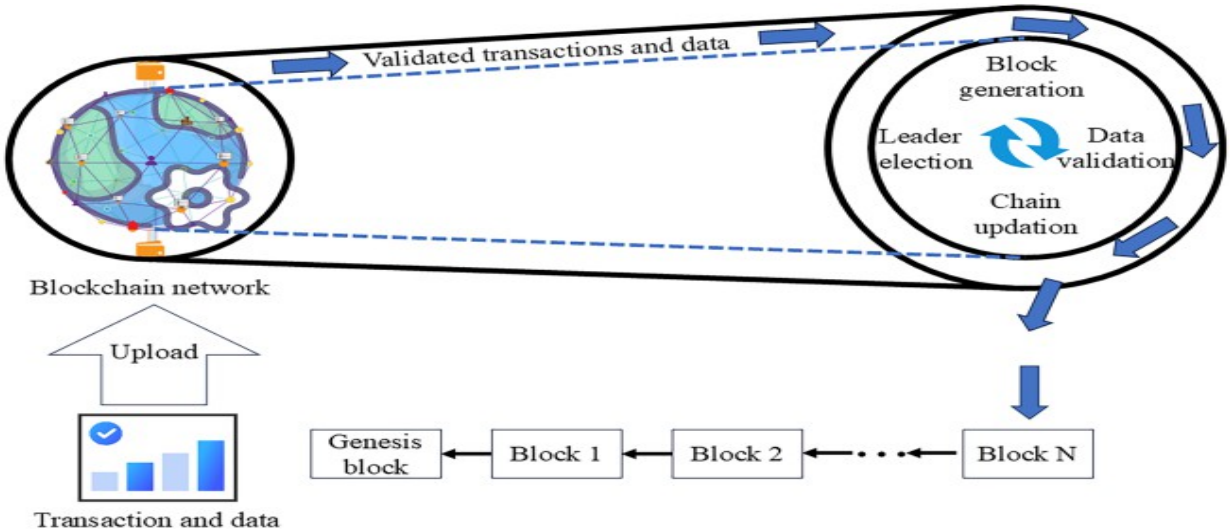



**Figure 5.** Consensus mechanism in the blockchain.

### 3.4. Sequential Transaction Processing

Another significant limitation comes from the sequential nature of the blockchain system. The blockchain system uses a chain of blocks, where each block contains a reference to the previous block through a cryptographic hash.

Because of this, transactions are processed sequentially, i.e., within each block. The system must ensure a consistent order for the transactions to guarantee deterministic state updates for all the nodes.

Sequential transaction processing limits the ability to process multiple transactions in parallel. The system, therefore, cannot take advantage of the current multi-core processors, which support parallel computing.

## 4. PERFORMANCE CHALLENGES

Blockchain technology has emerged as one of the most important innovations in modern distributed computing system. It provides a decentralized and secure method for storing and verifying digital transactions without relying on central authority. In a blockchain network, multiple nodes participate in maintaining a distributed ledger where transactions are recorded in blocks and linked together using cryptographic hashing.

One of the most widely known blockchain system is Bitcoin, which was introduced in 2008 as decentralized digital currency system. Bitcoin allows user to transfer digital assets directly without the need for intermediaries such as banks or financial institutions. While Bitcoin successfully demonstrated the feasibility of decentralized financial system, it also revealed significant limitations related to scalability and performance.

Scalability refers to the ability of a system to handle an increasing number of transactions, users, or workloads without reducing its performance. In the context of blockchain system, scalability means the capability of the network to process a large number of transactions efficiently as the network grows.

Performance in blockchain system is usually evaluated based on how quickly and efficiently transactions are processed and confirmed. A blockchain network must be able to process transactions quickly while maintaining important properties such as security, decentralization, and data integrity.

However, public blockchain system such as Bitcoin face serious scalability challenges. As the number of users and transactions increases, the network struggles to process transactions efficiently. This limitation is mainly due to the design principle used in the blockchain architecture.

For example, the bitcoin network can process approximately **3 to 7 transaction per second (TPS).** In contrast, traditional centralized payment system such as visa can process thousands of transactions per second. This large difference in transaction processing capacity highlights a significant performance gap between blockchain systems and traditional financial infrastructures.

As blockchain technology continues to gain popularity in areas such as digital payments, supply chain management, and decentralized finance, the ability to handle large transaction volumes become critically important. If blockchain system cannot scale effectively, they will struggle to support real-world applications that require high-speed transaction processing.

Therefore, scalability and performance limitations in Bitcoin blockchain systems represent a major research problem that must be addressed to enable the widespread adoption of decentralized technologies.

In order to clearly understand the scalability and performance challenges of the blockchain system, it is important to first examine the key performance metrics that researchers commonly used to evaluate how well a blockchain network operates. These metrics provide a structured way to measure the efficiency, responsiveness, and reliability of the transaction processing within a blockchain system.

Among the most important metrics used in blockchain research are throughput, latency, and finality. Each of these factors plays a significant role in determining how effectively a blockchain network can support a growing number of users and transactions.

## 1. Throughput

Throughput refers to the number of transactions that a blockchain network can successfully process within one second. It is typically measures using **transactions per second (TPS)**. This metric is widely used to evaluate how efficiently a system handles transaction workloads.

A blockchain network with higher throughput can process a larger number of transactions within a short period of time. This makes the system more capable of supporting large user basses and high transaction volumes. On the other hand, when throughput is limited, the network may struggle to keep up with incoming transactions.

In situations where the number of submitted exceeds the processing capacity of the network, transaction begin to accumulate in waiting areas known as the mempool. The mempool temporarily stores unconfirmed transactions until they are selected by miners and included in newly created blocks.

Bitcoin is widely known to have relatively low throughput compared with traditional payment systems such as credit card networks. This limitation restricts the number of transactions that can be processed within a given time frame. As result, during periods of high demand, users may experience delays and higher transaction fees. These limitations highlight one major scalability challenges faced by the bitcoin.

## 2. Latency

Latency represents the amount of time required for transaction to move from the moment it is submitted by a user to point where it becomes confirmed in the blockchain. In simple terms, Latency measures the delay between sending a transaction and having it recorded in the distributed ledger.

In bitcoin network, block creation time is 10 minutes. When a user submits a transaction, it must wait until a miner to select it from mempool and includes it in block. If transaction is not included in the next block, it must wait for a future block to created.

Even after a transaction is included in a block, most users and service providers do not immediately consider the transaction fully secure. Instead, they usually wait for additional blocks to added on top of the block containing the transaction. These additional blocks are known as **confirmations, and** they provide greater assurance that the transaction cannot be reversed.

Because of these factors, the effective waiting time for a transaction can be significantly longer than the initial block creation time. High latency can reduce the practically of blockchain systems, especially in applications that require fast and real-time transaction processing, such as retail payments or online services.

## 3. Finality

Finality refers to the point at which a transaction becomes permanently recorded in the blockchain and can no longer be altered or reversed. Achieving finality is an important aspect of maintaining trust and reliability in decentralized systems.

In the bitcoin network, transactions do not achieve immediate finality. Instead, they gradually become more secure as additional blocks are added after the block containing the transaction. This process reduces the risk of blockchain reorganizations or double-spending attacks.

A commonly accepted rule in the bitcoin ecosystem is to wait for six block confirmations before considering a transaction fully finalized. Since each block takes approximately ten minutes considering a transaction fully finalized. Since each block takes approximately ten minutes to generate, reaching this level of confirmation typically requires around one hour.

Although this confirmation process strengthens the security of the network, it also increases the time users must wait before a transaction is considered final. Therefore, finality directly affects the user experience and overall performance of the blockchain system.

Figure 1: Basic transaction processing workflow in a blockchain network.

5. **Solutions**

- **Segregated Witness (SegWit) in Bitcoin Blockchain Systems**

**1. Introduction**

The increasing use of the Bitcoin network has brought about various challenges to the network's performance. As the number of users grew, the network experienced increased congestion, slow confirmation of transactions, and increased costs of transactions. The major cause of these issues is the limited size of the blocks and the way transactions are stored within the blocks.

In the original design of the Bitcoin network, every transaction consists of various components such as input data, output data, and digital signatures used to verify transactions. The digital signatures take up a significant amount of space within the transactions. The blocks are of limited size, hence limiting the number of transactions that can be stored within the blocks. As the number of transactions increased, the network became slow and expensive to use.

To solve these problems without altering the original design of the network, the network was improved using the Segregated Witness protocol upgrade. The upgrade modifies the way transactions are carried out within the network. The upgrade separates the signature of the transactions from the other parts of the transactions. This modification enhances the network's capacity to process transactions.

**2. Background of SegWit Development**

The Segregated Witness was created to solve the problems of capacity and malleability in the Bitcoin network. Transaction malleability refers to the phenomenon where the identifier of a transaction, referred to as the transaction ID, can be altered prior to confirmation. This can pose problems in situations where the identifier of the transaction is required.

Segregated Witness was first proposed by Pieter Wuille, who was a developer of the Bitcoin network. It was first formally proposed in 2015 in the form of Bitcoin Improvement Proposal (BIP) 141. It was later activated in the network in August 2017 in the form of a soft fork, which means that nodes running older versions of the network can still operate without needing to upgrade.

The main reason for the development of Segregated Witness was to make the Bitcoin network more efficient without increasing the block size. It achieves this by reorganizing the way in which information in transactions is stored.

**1. Traditional Bitcoin Block Structure**

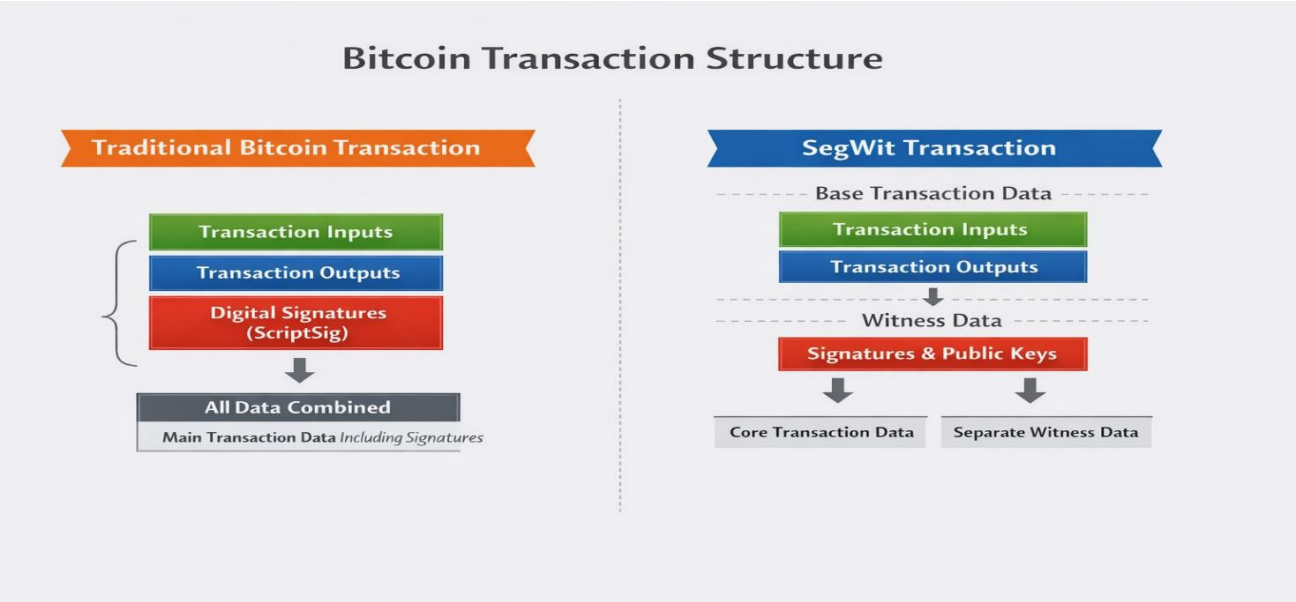
In the Bitcoin blockchain, before the introduction of Segregated Witness (SegWit), the transactions were stored in one data structure in each block. In each transaction, inputs, outputs, and digital signatures were combined in the transaction data. Digital signatures, which verify the ownership of the Bitcoins and authenticate the transaction, were stored in the transaction script (ScriptSig). This used a lot of space in each transaction, as it was a major part of the transaction. Since the blocks in the Bitcoin network were of 1 MB, this data structure limited the number of transactions that could be accommodated in each block.

```
Block
├── Block Header
└── Transactions
    ├── Inputs
    ├── Outputs
    └── Signatures (Witness data)
```

**2. Seg Wit Block Structure**

In Segregated Witness, also referred to as SegWit, the witness data used in the digital signature process is separated from the main transaction structure. In traditional transactions, the signature information was stored in the transaction input script. In Segregated Witness, the signature information is removed from the main transaction structure and placed in a separate section referred to as the witness field. In the main transaction section, information such as inputs, outputs, and amounts of transactions will still be stored. In the witness section, signature information and public keys will be stored. By separating the signature information from the main transaction structure, the space taken in the block will reduce, thus increasing the number of transactions that can be accommodated in a single block.

```
Block
├── Block Header
├── Transactions (base data)
└── Witness Data (separate section)
```




**Quantitative Proof of SegWit Scalability Using Block Weight**

A big change brought by Segregated Witness (SegWit) on Bitcoin was how block capacity gets measured, swapping out the old rigid size limit. Rather than capping each block at exactly 1 MB, it now uses something named block weight - this tweak opens up smarter ways to fill space. Because of this shift, data fits together differently, making room without breaking past limits. Efficiency climbs since what counts toward block size now depends on a weighted system instead of raw volume.

Weight comes from how much data fits inside a Bitcoin block. A special math rule decides the total number based on parts packed into it. Each piece adds up differently depending on type. Some count more than others toward the final score

$$\text{Block Weight} = (\text{Base Transaction Size} \times 4) + \text{Witness Size}$$

Here's what makes up the transaction. The basic details - like where it came from, where it goes, and its ID - form the foundation. What backs it up comes next: proof through public keys and electronic marks that confirm authenticity. These pieces fit together, one following the other without overlap.

Weight values shift between kinds of data under SegWit. Every byte of regular transaction details takes up four weight points instead. For every byte tied to proof info though, just a single point applies. That makes the proof section count far less - only 25 percent as much by comparison.

One thing limits how heavy a block can get on Bitcoin's system

4,000,000 weight units

This setup uses math to fit extra transaction details into each block, even though the old 1 MB rule still stands. Since proof information takes up less space under the new counting method, more overall data can squeeze in when transactions follow the SegWit format.

Block space gets stretched when this system runs. With more SegWit deals inside, one chunk might grow to nearly double what old blocks allowed - hitting up to 2 megabytes instead of just 1. Size shifts based on how many modern transfers fill it.

So, here's how it goes: the block weight math shows, in clear numbers, just why adding SegWit lifts limits on transactions while also helping Bitcoin handle growth more smoothly over time.

**5. SegWit Adoption in the Bitcoin Network**

One day after another since 2017, more wallets began using SegWit without much noise. Over months, exchanges followed - drawn by smaller fees and smoother processing. Bit by bit, recognition grew; advantages became harder to ignore. Efficiency started mattering more, so integration spread through big-name platforms. Transaction speed changed quietly, yet noticeably, under the surface.

**Limitations of SegWit**

Even though SegWit makes Bitcoin's blockchain work better, scaling issues remain unsolved. Block space gets stretched, yet the core structure stays unchanged. Because of this design choice, more transactions fit - but only by a bit. When lots of people send at once, delays can still happen. The system breathes easier, just not enough to avoid jams.

Not everyone using SegWit holds back its impact. When wallets, exchanges, or people stick to old transaction styles, most activity stays in the original form. Block space gains then remain limited. Full change only happens if usage spreads widely.

Still, SegWit doesn’t’t push Bitcoin’s transaction speed much higher when set beside today’s mainstream payment platforms. Though SegWit is in play, the number of transactions each second stays well under what big corporate networks handle regularly.

Only after fixing how data moves inside blocks did things get quicker. Still, handling way more users means reaching beyond basic upgrades. One fix alone won’t stretch far enough when demand grows wild. So even if this change helps now, stacking extra layers later makes the system last longer. Without mixing tools built above, heavy traffic would overwhelm everything eventually.

**Conclusion**

One big change on Bitcoin came with Segregated Witness. Instead of mixing signatures into regular transaction parts, they moved them outside, freeing up room. This shift lets more deals fit inside every block. Because of better space use, the network handles growth a bit easier now. Fixing transaction IDs so they cannot be tweaked was another result. That stability opened doors for tools like the Lightning Network. While problems around size limits still exist, this upgrade helped push things forward. Progress in how chains work got a quiet boost here.

## 2.Lightning Network Layer Two Scaling

### 2.1.Lightning Network explained

Built above Bitcoin's main chain, the Lightning Network helps fix slow transfers and steep costs. Though it started as one idea, now it handles more users without clogging up. Fees drop because money moves faster between people directly. Instead of waiting minutes, payments zip through almost instantly. Scalability improves when traffic avoids the base layer. This second-layer approach keeps Bitcoin working smoothly under pressure. Problems like delays fade when transactions shift off-chain.

Off the beaten path from Layer 1, the Lightning Network handles trades without touching the main chain each time. Most moves happen behind the scenes, unseen by the ledger. When things settle, just one update hits the blockchain - like a summary after whispers are done. The bulk stays out of sight until it matters.

With this method, speed gets a boost without putting Bitcoin's safety at risk. Performance jumps up even as protection stays strong. Speed increases greatly but defense never drops. The system runs faster yet keeps its guard tight. Gains in efficiency happen alongside solid safeguards.

### 2.2.Lightning Network addresses slow bitcoin transaction speeds

Bitcoin has the following limitations:

- Block size is limited (~1 MB)
- Every block takes about ten minutes to form
- Just seven transactions every second get handled. That is how fast it moves. Speed does not go beyond that mark. Each second passes with the same limit. Performance stays fixed at this point

Because of such constraints:

- Transactions become slow during high demand
- Transaction fees increase
- Network congestion occurs

Speedy fixes came through a new system built for cheaper payments. Offshoots of blockchain tech made delays shrink overnight.

### 2.3.How the Lightning Network Operates

A single connection links two people, letting money move instantly. These pathways open only when needed, staying quiet otherwise. When someone sends value, it flows through their private route. No middle steps slow things down - just quick transfers from one side to another.

Step 1: Opening a Payment Channel

- A pair of people set up a shared wallet requiring both signatures. Inside this setup, they place a quantity of Bitcoin. One person alone cannot move the funds - both must agree. Their joint control starts once the coins are inside.
- Frozen into digital stone, this exchange lives on the blockchain. A permanent mark sits where data never fades. Stored across machines, it resists change. What happened stays written, visible yet untouched by time.

Step 2: Off Chain Transactions Second Phase

- Faster messages fly between people who trade often. One after another, deals move without delays piling up. Each exchange pops through right away when sent both ways.
- Off the chain, these deals stay unlogged.
- Now the scale shifts between them again. Balance changes hands without warning. One gains what the other quietly loses.

Step 3: Closing the Channel

- Fresh after a transaction wraps up, the updated total lands permanently on the blockchain.
- Just two trades show up on the list Opening transaction Closing transaction
    - Opening transaction
    - Closing transaction

### 2.4.Lightning Network in Action

Suppose :

One person puts in a bitcoin; another does the same to start things off together. Each of them adds one whole bitcoin when setting up their shared space.

Transactions :

- A sends 0.2 BTC to B
- B sends 0.1 BTC to A
- B sends 0.3 BTC straight to A. Payment moves without delay between them

A single updated amount appears on the chain, not three separate entries. The network logs just what ends up owed, skipping every step before it.

Final state:

- A equals zero-point six bitcoin
- B equals one point four Bitcoin

Fewer demands hit the network

now.

**2.5.Lightning Network Speeds Transactions Low Fees Scalability Off Chain Payments Instant Settlement**

1. Instant Transactions

Sending money takes just seconds, so there is no need to pause for blocks to confirm. Speed comes built in, meaning everything moves before delays can catch up.

2. Low Transaction Fees

Fees stay tiny since transaction data skips the blockchain entirely.

3. Scalability Improvement

Faster processing happens when deals move away from the main chain.

4. Microtransactions Support

5. Decentralized Network

Tiny Bitcoin transfers become possible here, something the main network struggles with. A single Satoshi moves without fuss, where the base layer would stall. This layer handles crumbs effortlessly, unlike its heavier counterpart. Transactions shrink to almost nothing, yet still get through cleanly. Minuscule values flow freely, bypassing the congestion above.

Folks running their own nodes keep things moving - no single boss calling the shots.

**2.6.Lightning Network Benefits**

- Reduces blockchain congestion
- Increases transaction speed
- Lowers transaction fees
- Enables real-time payments
- Improves scalability of Bitcoin

### 2.7.Limits of the lightning network

Lightning Network benefits and drawbacks

1. Channel Management Complexity

Fiddling with channel settings might feel tricky at first. Getting in and out takes a few steps that aren't always clear.

2. Liquidity Issues

Transactions need space in the channel to move through. Enough room keeps things flowing without stopping.

3. Offline Requirement

Fraud stays low when people keep an eye on the channels they use. Watching closely helps catch problems before they spread through the system. It is regular attention that stops small issues becoming big ones down the line.

4. Routing Problems

A wrong turn might happen when tracing steps from one person to another. Sometimes the connection just does not show up.

5. Security Concerns

Few people stay alert all the time, so safety can slip when attention fades.

### 2.8.Lightning Network Enables Faster Bitcoin Transactions

The Lightning Network makes Bitcoin handle more transactions faster

- Moving transactions off-chain
- Fewer trades happen directly on the primary chain. That shifts activity elsewhere. Instead, things move off to separate systems. This eases pressure on the core network. Less load means smoother operation overall
- Increasing effective transactions per second (TPS)

Bitcoin can process thousands up to millions of transactions every second this way, while Layer 1 manages just around seven.

### 2.9.Conclusion

Off the main blockchain, transactions move quicker when using payment channels. Congestion drops because activity shifts away from Bitcoin's base layer. Fees shrink since less space on chain means lower demand. Speed

improves dramatically under normal conditions. This approach tackles scaling issues without changing core rules.

Still, problems pop up - like juggling channels or running short on funds. Even with those hiccups, the system boosts how fast and smoothly Bitcoin works when people actually use it.

## 6. Discussion

Looking at how Bitcoin handles growth reveals a core tension among three things: staying decentralized, keeping data safe, leaving room to scale up - a mix people call the blockchain trilemma. Built first to protect safety and spread control widely, its structure now limits how many transactions move through each second. Not handling big loads well comes down to capped block space, slow check times, plus lag across nodes. When more people join, traffic backs up - fees rise, waits stretch out. That drags on usability. Global payments? Tough fit when things bog down this much.

Now comes Segregated Witness, known as SegWit, built to tackle growing pains in transaction handling. It rearranges storage of transaction details, squeezing more deals into each block - no need to touch the cap on block size. Meanwhile, another path opens: the Lightning Network runs trades outside the main chain, letting people swap value fast, again and again, while spending less. What started as bottlenecks now bends into speed.

Still, every fix has its downsides. Not everyone uses SegWit, so it can’t do everything it was meant to. The Lightning Network brings trickier steps - people must learn how payment channels work and handle funds carefully. Security questions pop up too, along with whether payments find their way smoothly across paths. Running outside the main chain raises doubts about steady performance over time.

Still, even as these fixes boost capacity, trust begins to gather around key points - like Lightning Network nodes - which nudges things less toward distributed control. So, weighing how such shifts might reshape blockchain's broader landscape becomes necessary over time.

Even so, gains have come slowly. Still, scaling holds back wider use. Work never stops because better ways must emerge - ways that handle growth without losing safety or ease. Progress depends on fresh thinking, each step forward shaped by trial, patience, time.

## 7. Conclusion

In this research paper, we closely examined the scalability and performance challenges of the Bitcoin blockchain system. The study pointed out key issues, such as the block size limit, slow transaction validation, and network delay. These factors contribute to lower efficiency and higher transaction costs.

The paper also explored contemporary solutions like Segregated Witness (SegWit) and the Lightning Network. These solutions aim to boost transaction speed and cut down on congestion. They show that it is possible to make scalability improvements without sacrificing the main principles of blockchain technology.

The paper also looked at modern solutions like Segregated Witness (SegWit) and the Lightning Network. These aim to increase transaction speed and lessen congestion. These solutions show that it is possible to improve scalability without fully sacrificing the main principles of blockchain technology.

However, it's clear that no single solution can completely tackle the scalability problem. Each method has its own pros and cons. So, a mix of on-chain and off-chain solutions, along with future technological improvements, will be needed to create a balanced and scalable blockchain system.

Overall, Bitcoin keeps evolving as new innovations appear. The future of blockchain scalability relies on continued research, broader use of current solutions, and the creation of new methods that can uphold security, decentralization, and performance at the same time.

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